What is in `embeddings.npz`?

Aviral Garg · Forgis screening task · 10 Oct 2026

Every number here comes out of `code/analyze.py` (numpy/scipy/torch, no sklearn, seed 0, about 4 minutes on CPU). Raw values are in `out/results.json`.

1. Identification

Given: `T` (2500, 512) f32, `I` (2500, 512) f32, `Y` (2500,) i64. No other metadata.

- `Y` is a class label: 31 distinct values in [1, 34], 31–33 absent ⇒ a subset of a 34-class dataset. 49–150 samples/class; majority class 6.0%.
- Not ℓ_2-normalised: $\|\texttt{T}\| = 9.06 \pm 0.86$, $\|\texttt{I}\| = 10.19 \pm 0.46$. Raw encoder output, taken before any normalisation layer.
- Two different encoders: per-dim variance correlation 0.0017, mean per-dim correlation 0.18, and only 4 of 512 dims reach $|r| > 0.5$. They share no basis.
- `T`'s inputs repeat exactly, `I`'s never do: `T` 2075/2500 unique (173 duplicate groups, max multiplicity 30), `I` 2500/2500. 98.8% of duplicate groups sit inside one class ⇒ `T`'s inputs are templated per class, `I`'s are per-instance.
- Row order carries nothing: $\cos(\texttt{I}_i, \texttt{I}_{i+1}) = 0.6579$ against 0.6581 for a random pair, and the same at lags 2, 5, 10 and 50. `Y` has 2416 runs in 2500 rows.

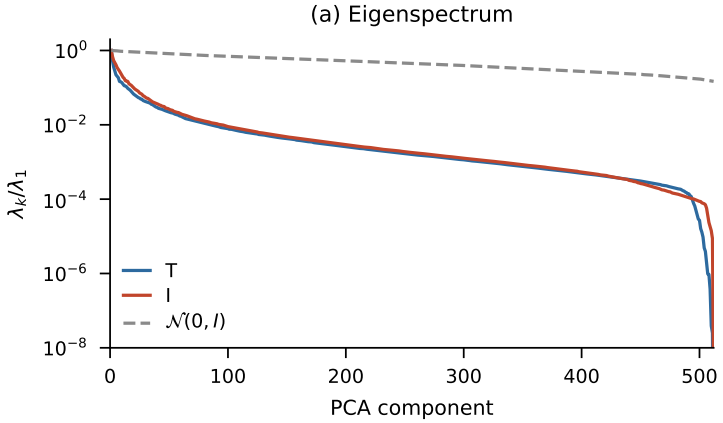
Hypothesis: a CLIP-style dual encoder, with `T`=text, `I`=image and `Y`=class.

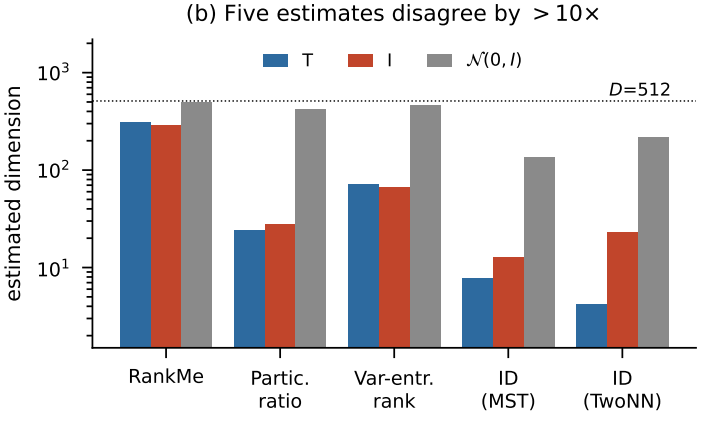
- For: D=512 is CLIP ViT-B/32's shared embedding width for both towers; the vectors are unnormalised, consistent with a dump taken before the ℓ_2 layer; two separate towers explain the disjoint bases; templated captions explain `T`'s duplicates; the large mean-offset (§3) matches CLIP's documented modality gap.
- Against / open: paired similarity barely exceeds same-class similarity (§3), so the row pairing is class-level; and 34 classes on their own weren't enough for me to pin down the source dataset. I'm putting this forward as a hypothesis with its evidence, and nothing stronger.
- Decisive test, which I could not run without the source data: encode known image/caption pairs with ViT-B/32 and compare norms, gap and spectrum. I have not measured reference CLIP norms myself.

It is not a HEPA dump. HEPA (Petersen et al., arXiv:2605.11130) emits a (h_t, h^*) pair per sample, so I tested for that. Three checks, all of them negative:

- D=512 ≠ HEPA's d=256 (their §3.1).
- No temporal structure, but HEPA's pairs index $(t, \Delta t)$ over episodes.
- HEPA pretrains with SIGReg, yet these sit at SIGReg distance 23.9 from $\mathcal{N}(0, I)$ vs. 0.57 for a matched Gaussian (§2).

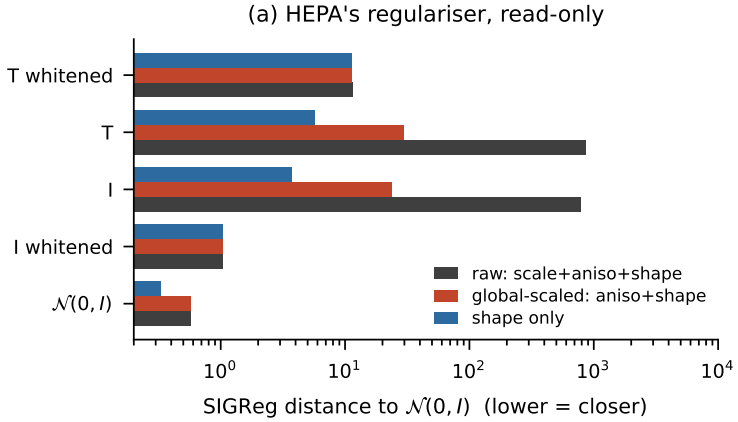
2. Geometry: every rank estimate disagrees

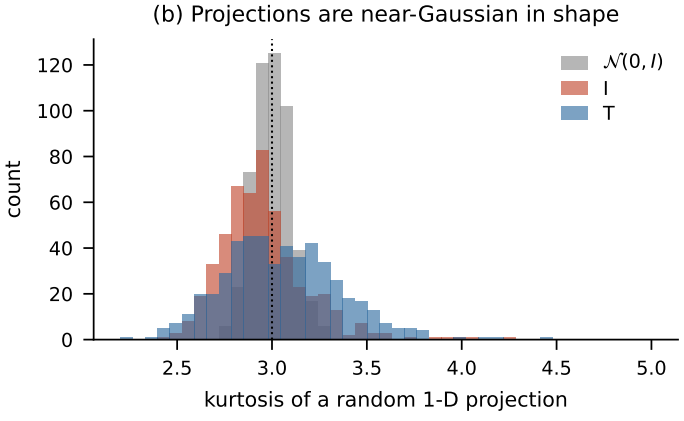




- Strongly anisotropic: top-10 of 512 dims hold only 12–15% of variance; 90% needs 110 (`T`) / 93 (`I`) dims; the spectrum then falls five orders of magnitude.
- Five standard estimates span > 10× on the same data (`T`/`I`): RankMe 309/292, participation ratio 24/28, variance-entropy rank 71/66, MST-ID 7.7/12.9, TwoNN-ID 4.3/23.2.
- They disagree on the ordering too: TwoNN says `T`≪`I` (4.3 vs 23.2), RankMe says `T`>`I` (309 vs 292).
- On an isotropic Gaussian control both ID estimators badly underestimate (136, 216 vs. a true 512) ⇒ at N=2500, D=512 absolute ID values are untrustworthy. So I only compare rank metrics within one estimator, and I name the estimator every time.

3. Isotropy, measured with HEPA's own regulariser

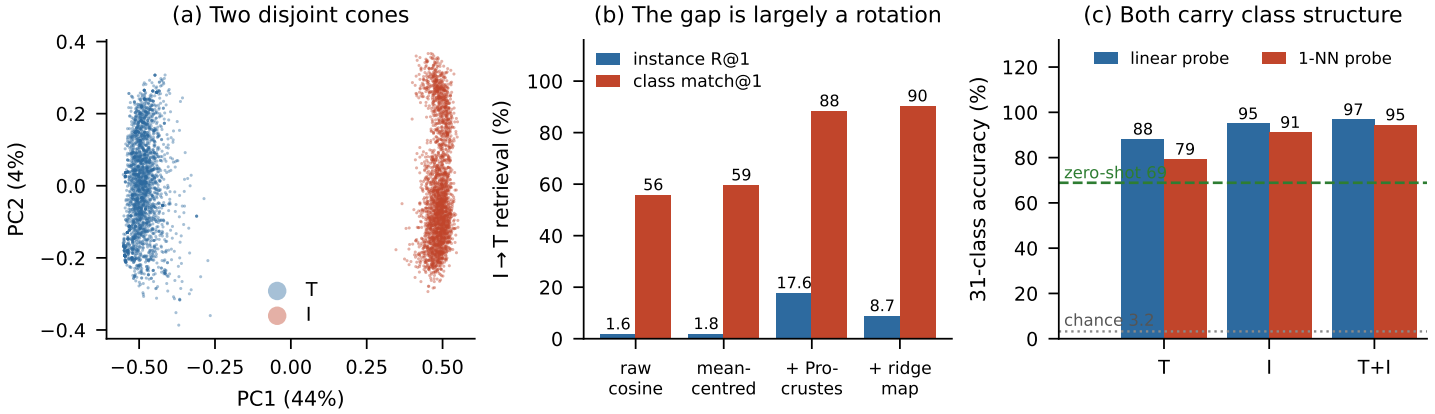




- SIGReg implemented exactly as LeJEPA algorithm 1 (Balestriero & LeCun, arXiv:2511.08544): random unit directions, unstandardised projections, Epps–Pulley with window $e^{-t^2/2}$, 17-knot trapezoid on $[-5, 5]$. Run read-only, decomposed three ways (`T` / `I` / $\mathcal{N}(0, I)$):
  – raw (scale+aniso+shape) 860 / 781 / 0.57 · global rescale (aniso+shape) 29.8 / 23.9 / 0.57 · per-projection standardised (shape only) 5.7 / 3.7 / 0.33.
- Almost all of the departure is scale and anisotropy; the shape is already close to Gaussian. Mean projection variance is 0.044 / 0.070 where SIGReg wants 1.0, while projection kurtosis is 3.06 / 2.95 against a Gaussian 3.00. This is what drives §4.

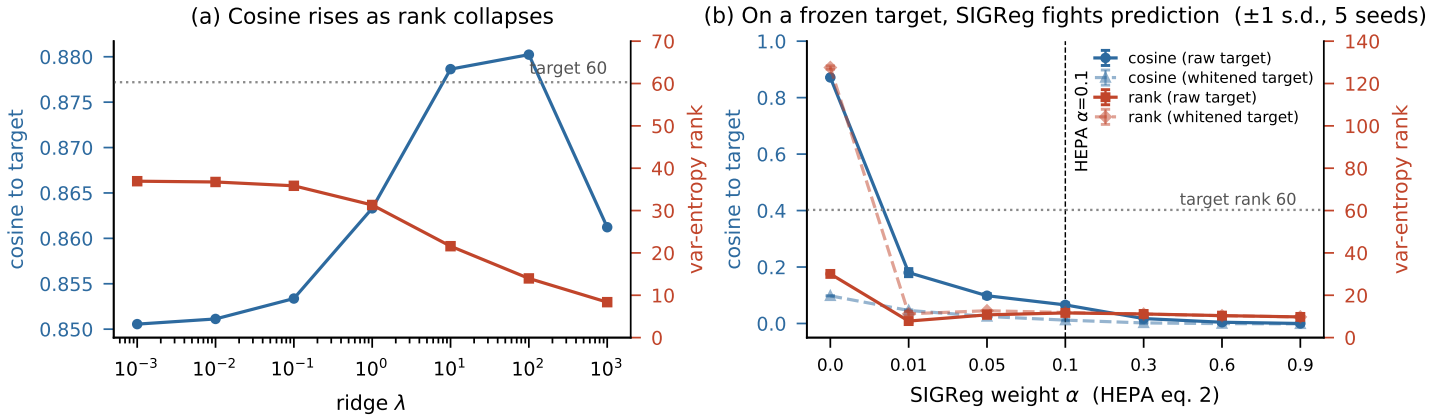
- Whitening T makes it worse: shape-only rises 5.7→11.4 and kurtosis 3.06→4.15, because whitening amplifies near-null directions into heavy tails. Whitening I behaves (1.04), so a blanket "whiten first" step isn't safe here.

4. The modality gap is largely a rotation



- Disjoint clouds: $\cos(\bar{\mathtt{T}}, \bar{\mathtt{I}}) = 0.319$, $\|\bar{\mathtt{T}} - \bar{\mathtt{I}}\| = 0.97$ (normalised). PC1 alone separates them.
- Raw cosine barely sees the pairing: paired 0.278, same-class 0.272, random 0.222. The 0.006 margin says the correspondence is class-level.
- One orthogonal Procrustes rotation recovers most of it (I → T): instance R@1 1.6%→17.6% (11×; chance 0.04%), R@5 7.2%→35.0%, class match@1 55.7%→88.2%. Mean-centring alone gives almost nothing (1.8% / 59.5%).
- Zero-shot (classify I by nearest class-mean T prototype): 68.9% → 90.1% after the same rotation; chance 3.2%.
- But not a pure rotation: an unconstrained ridge map wins on class match (90.2%) and loses on instance R@1 (8.7%), with singular values far from flat (mean 0.23, s.d. 0.35).
- Both carry class structure: linear probe 88.2 / 95.0 / 96.8% and 1-NN 79.3 / 91.2 / 94.5% (T / I / concat). The linear−kNN gap is +8.9 pts on T but +3.8 on I, so T's class information is linearly decodable while being less metrically organised. Templated inputs would do exactly that.

5. Prediction in embedding space: cosine lies, rank tells



- Setup: the obvious JEPA experiment here. Context T, target I, predict in representation space. 80/20 split, held-out metrics, and MLP results are mean ± s.d. over 5 seeds.
- A ridge map looks excellent while quietly collapsing. Over $\lambda : 10^{-3} \to 10^{2}$, cosine rises 0.851→0.880 and R^2 0.16→0.35, while the prediction's variance-entropy rank falls 36.9→14.0 against a target rank of 60.3. Cosine and rank move in opposite directions across the whole useful range of λ. Had I reported cosine on its own, it would have hidden a 4× collapse. Retrieval tracks the rank: R@1 peaks at 9.6%, then falls to 3.4% at λ=10^3.
- SIGReg on the predictor does not fix it here. 2-layer MLP under HEPA eq. (2), $\mathcal{L} = (1-\alpha)\|\hat{h} - h^*\|_1 + \alpha\mathcal{L}_{\text{SIGReg}}$: at α=0, cosine 0.8715±0.0003 / rank 30.1±0.2; at HEPA's own α=0.1, cosine 0.066±0.007 / rank 11.6±0.3. Every $\alpha \geq 0.01$ destroys prediction, and the seed-to-seed spread is far smaller than the effect.
- Part of the reason is a scale conflict. SIGReg pins the output to $\mathcal{N}(0, I_d)$, so unit covariance, while the frozen target has mean projection variance 0.070. The two terms pull toward covariances that can't both hold. Control: against a whitened target (projection variance 1.000) that conflict is gone, and prediction still collapses (cosine 0.0979±0.0006 → 0.046±0.002 at α=0.01; rank 127.6±0.8 → 11.2±1.1). So the scale conflict is part of it, but it isn't the whole story.
- What this says about HEPA. HEPA trains context and target encoders jointly (§3.1: "both encoders are trained jointly via the optimizer"), so its target is free to migrate toward the isotropic Gaussian SIGReg demands. On a frozen target that freedom is gone, and no α buys both prediction and rank. So joint training isn't a convenience in HEPA; SIGReg needs it. That is where my proposal starts.

6. Caveats

- MLP results are 5 seeds (mean ± s.d.) and the ridge sweep is closed-form and exact. But there is one split throughout: I don't resample the 80/20 partition, so the split is still an untested source of variance.
- The CLIP identification is unconfirmed (test named in §1).
- The α sweep is one architecture at one budget (2-layer MLP, 200 epochs). I'd stand behind the direction, that SIGReg and the prediction term conflict on a frozen target. I wouldn't stand behind the exact crossover point.
- Next: resample the split; where Procrustes fails per class (the residual 12% is likely concentrated in the duplicated-caption classes); whether the optimal map's non-orthogonality is itself class-structured.